**Lamarckian Replicators in Darwinian Hierarchies: Reconciling Gene-Centered and Multilevel Selection Approaches to Legal Evolution**

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# Abstract

Legal evolution presents a theoretical puzzle that neither gene-centered nor multilevel selection frameworks, taken independently, can resolve. This Article argues that legal systems evolve through a hierarchical selection process where the Lamarckian character of cultural transmission, far from collapsing the hierarchy into a single level, generates a partial decoupling between three analytically distinct tiers of evolutionary dynamics. At the *micro* level, judicial interpretation and doctrinal adjustment operate through Heteronomous Bayesian Updating (HBU), a mechanism of Lamarckian inheritance where legal professionals learn norms by observing institutional reactions rather than substantive outcomes. At the *meso* level, competition among doctrinal memeplexes produces patterns structurally analogous to species selection as described by Stephen Jay Gould and Niles Eldredge: normative bodies originate, persist in stasis, and go extinct, with long-term trends emerging from differential proliferation rates rather than gradual transformation. At the *macro* level, constitutional architecture generates basins of attraction whose depth, measured by the Constitutional Lock-in Index (CLI), determines the duration of institutional stasis and the threshold for punctuated change.

The central contribution is to argue that the apparent tension between Richard Dawkins' replicator-centered framework and Gould's hierarchical selection theory dissolves in the domain of cultural evolution, precisely because the Weismann barrier that makes the biological debate intractable does not exist for memes. Legal norms are simultaneously faithful replicators and genuine Darwinian individuals at multiple hierarchical levels. Drawing on Mario Bunge's systemist ontology, Daniel Dennett's Design Space, Peter Godfrey-Smith's criteria for paradigm Darwinian populations, Joseph Henrich's gene-culture coevolution framework, and Telmo Pievani's analysis of contingency and exaptation, this Article proposes an integrated model that preserves the explanatory power of Extended Phenotype Theory (EPT) while incorporating the genuine insights of multilevel selection. The model generates three falsifiable predictions supported by preliminary evidence from four jurisdictions. IusSpace, a twelve-dimensional framework for mapping legal evolution previously developed within this research program, provides the environmental constraint topology within which the hierarchical selection operates.

**Keywords:** hierarchical selection, multilevel selection, extended phenotype, legal evolution, Lamarckian inheritance, memes, punctuated equilibrium, constitutional lock-in, evolutionary game theory, Darwinian jurisprudence, systemism, IusSpace, exaptation, gene-culture coevolution

## I. Introduction: A Debate That Should Not Exist

Evolutionary biology's longest-running theoretical dispute concerns the level at which natural selection operates. For over four decades, two formidable intellectual traditions have offered competing answers. Richard Dawkins, building on the gene-centered perspective consolidated in The Selfish Gene (1976) and extended in The Extended Phenotype (1982), argues that genes are the fundamental units of selection: replicators whose differential survival explains all evolutionary phenomena, with organisms serving merely as vehicles constructed by cooperative gene cartels. Stephen Jay Gould, in The Structure of Evolutionary Theory (2002), proposes instead a hierarchical model where selection operates simultaneously at six levels, from genes through cell lineages, organisms, demes, species, and clades, with species selection constituting a genuinely irreducible macroevolutionary force.

The dispute has proven remarkably resistant to resolution in biology, in part because the Weismann barrier, the developmental separation between germline and soma that prevents acquired somatic modifications from entering the hereditary material, makes it extraordinarily difficult to disentangle the causal contributions of different hierarchical levels. If phenotypic changes cannot be transmitted directly, the gene remains the only entity with the longevity, fecundity, and copying-fidelity required of a true replicator. Gould's interactors, whatever their causal role in determining selective outcomes, must ultimately cash out in terms of gene frequencies.

This Article argues that the Dawkins-Gould dispute should not exist in the domain of legal evolution, and that the reason it should not exist suggests something fundamental about the structure of institutional change. The argument proceeds from a simple but consequential premise: legal norms, understood as cultural replicators or memes, operate in a transmission environment where the Weismann barrier is absent. Legal practitioners can and routinely do modify doctrines during the very act of transmitting them. A judge who applies a precedent simultaneously reproduces it and alters it; the alteration enters the hereditary stream immediately, without any barrier between the 'somatic' act of application and the 'germinal' act of transmission. Cultural inheritance is, as both Dawkins and Gould have acknowledged, fundamentally Lamarckian.

The absence of the Weismann barrier does not, however, collapse the evolutionary hierarchy into a single level. This is the non-obvious claim at the center of this Article. One might expect that if acquired modifications transmit directly, everything changes fast, and there is no basis for stable higher-level entities to emerge as units of selection. The empirical reality of legal systems contradicts this expectation. Constitutional architectures persist for centuries despite continuous doctrinal modification at lower levels. Argentina's labor law framework has resisted 23 reform attempts since 1991, exhibiting a Constitutional Lock-in Index of 0.89, even as judicial interpretation of specific provisions shifts constantly. The micro level is Lamarckian and fluid; the macro level is punctuated and rigid. The coexistence of these two patterns within a single system is the phenomenon that demands explanation.

The explanation I propose is that the Lamarckian character of cultural transmission, rather than eliminating hierarchical structure, generates a specific form of partial decoupling between levels. Rapid adaptation at the micro level does not translate into rapid change at the macro level because constitutional architectures are emergent properties of the institutional system, not aggregations of individual norms. They exhibit what Gould, following Elizabeth Lloyd, termed emergent fitness: their persistence or extinction depends on system-level properties that cannot be decomposed into the fitness of any individual norm. The depth of the basin of attraction that holds a constitutional order in place is a macro-level quantity, just as geographic range or population structure are species-level quantities in biological evolution.

This integrated model draws on six intellectual traditions that, despite their apparent diversity, converge on a common insight: complex adaptive systems exhibit level-specific dynamics that require level-specific explanatory frameworks. From Dawkins, I retain the norm-as-replicator foundation and the extended phenotype concept. From Gould and Eldredge, I adopt the hierarchical structure, emergent fitness, and punctuated equilibria. From Mario Bunge, I borrow the systemist ontology and the CESM model. From Daniel Dennett, I take Design Space, the universality of the Darwinian algorithm, and the principle of competence without comprehension. From Peter Godfrey-Smith, I adopt the criteria for distinguishing paradigm Darwinian populations from marginal ones and his challenge to naive memetics. From Joseph Henrich, I incorporate gene-culture coevolution and the WEIRD/non-WEIRD institutional bifurcation. And from Telmo Pievani, I take the contingency perspective and the exaptation concept that illuminate institutional repurposing.

The synthesis attempted here is not merely additive. Placing these six traditions in conversation generates insights that none of them produces independently. Dawkins' framework alone cannot explain why micro-level plasticity coexists with macro-level rigidity, because it predicts that selection at the replicator level should produce uniform change rates across all levels. Gould's framework alone cannot explain why legal hierarchies exhibit such sharp level differentiation in a Lamarckian transmission environment, because his biological model assumes Weismannian inheritance at the base level. Bunge's systemism alone provides the ontological scaffolding for levels but lacks the evolutionary mechanism to explain the dynamics within and between them. Dennett's universal Darwinism alone establishes the algorithm's applicability but does not specify how the algorithm's operation differs across levels. Godfrey-Smith alone provides the criteria for evaluating Darwinian populations but does not develop a positive model of hierarchical cultural evolution. Henrich alone explains the cultural psychology underlying norm adherence but does not connect it to the formal structure of constitutional architecture. Only the integration of all six traditions produces a model that simultaneously explains micro-level plasticity, meso-level regime persistence, macro-level constitutional stasis, and the systematic variation in these patterns across jurisdictions.

## II. Theoretical Foundations: The Dawkins-Gould Debate and Its Resolution in Cultural Evolution

### A. The Replicator-Vehicle Framework

Dawkins' framework rests on a fundamental ontological distinction between replicators and vehicles. A replicator is any entity capable of producing copies of itself with sufficient fidelity to sustain cumulative selection. Its evolutionary success depends on three properties: longevity (persistence across time in the form of copies), fecundity (rate of copying), and copying-fidelity (accuracy of replication). In biological evolution, genes satisfy these criteria. Organisms, by contrast, are vehicles: survival machines constructed by cooperative gene cartels to propagate the replicators housed within them. Vehicles interact with the environment and compete for resources, but they do not replicate themselves in the relevant sense.

The extended phenotype concept generalizes the vehicle idea. Dawkins observed that genes can influence the world beyond the organism's body: a beaver's dam, a caddisfly larva's case, a parasite's modification of its host's behavior are all phenotypic expressions of genes, no less than eye color or bone structure. The extended phenotype dissolves the organism's boundary as a privileged unit of analysis, revealing that the true unit of selection can extend its causal influence arbitrarily far into the external environment.

When Dawkins introduced memes as cultural replicators, he extended this logic to human culture. Memes propagate in the 'soup of human culture' by leaping from brain to brain through imitation. Like genes, their success depends on longevity, fecundity, and copying-fidelity. Like genes, memes build extended phenotypes: the institutions, practices, and material artifacts that serve as their reproductive machinery. Dawkins was explicit that memes operate independently of genetic advantage; once the human brain evolved the capacity for rapid imitation, memes began evolving for their own benefit, sometimes in ways neutral or detrimental to their hosts' genetic fitness.

### B. Gould's Hierarchical Alternative

Gould challenged the gene-centered framework on two fronts. First, he argued that Dawkins confused the accounting of evolution (tracking gene frequencies) with the causality of selection (the interaction between phenotypes and environments). David Hull formalized this distinction through the concept of the interactor: an entity that interacts with its environment as a coherent whole, producing differential replication. For Gould, causality resides in interaction, not mere replication; the gene may be the bookkeeper, but the organism is the causal agent.

Second, Gould proposed that the interactor concept applies at multiple levels simultaneously. The theory of punctuated equilibrium, developed with Niles Eldredge in 1972, provided the empirical foundation. If species originate rapidly in geological time, persist without significant directional change for millions of years (stasis), and then go extinct as discrete entities, they satisfy the criteria for Darwinian individuality. Species selection becomes possible when long-term trends result from differential speciation and extinction rates among stable species within a clade, rather than from the gradual transformation of organisms within a single lineage.

Gould's mature framework identifies six levels: genes, cell lineages, organisms, demes, species, and clades. He insisted that the hierarchy is not fractal; the dominant forces vary by level. Random drift dominates at the genetic level. Classical natural selection dominates at the organismal level. Exaptation plays a proportionally larger role at the species level. The observable patterns of the history of life represent the complex, sometimes conflicting outcome of selection operating across all levels simultaneously.

The concept of emergent fitness, developed with Elizabeth Lloyd, provided the criterion for genuine species-level selection. Initially, Gould and Vrba required emergent characters: properties arising from non-linear interactions among constituents that do not exist at lower levels. Later, Gould relaxed this to emergent fitness: a trait need not be emergent as a character; what matters is that it confers fitness advantages at the species level irreducible to the reproductive success of individual organisms.

Each level in Gould's hierarchy exhibits its own characteristic dynamics, and the differences between levels are not merely quantitative but qualitative. At the gene level, phenomena like gene duplication and 'selfish DNA' demonstrate that genetic elements can proliferate within a genome independently of the organism's overall adaptive needs. When this cell-level competition is suppressed by organism-level integration mechanisms, the multicellular body functions as a coherent Darwinian individual; when suppression fails, the result is cancer, the pathological reassertion of cell-level selection against organismal interests. At the deme level, interdemic or group selection, long controversial but now widely acknowledged as at least occasionally significant, operates on local populations and is likely crucial for the evolution of cooperative and altruistic behaviors that individual selection struggles to explain. At the clade level, the highest tier, selection is likely weak because clades lack the cohesive 'glue' that binds lower-level entities: they are definable only retrospectively and contain relatively few parts. This asymmetry across levels, with the organismal and species levels carrying the heaviest explanatory burden, is a feature of the model, not a limitation.

The cross-level interactions that Gould identifies are particularly relevant for the legal analogy. Selection at one level can interact with adjacent levels synergistically, orthogonally, or in direct opposition. Upward causation, in Vrba's terminology the 'effect hypothesis,' occurs when organism-level adaptations cascade upward to influence species-level outcomes: a species proliferates not because of its emergent properties but as a byproduct of its organisms' individual fitness. Downward causation operates in the reverse direction: when a species or clade succeeds, it automatically expands the frequency of all genes and organisms nested within it, regardless of whether those lower-level entities were directly selected. The observable evolutionary pattern at any moment is a compromise, often a conflicting one, produced by the simultaneous push and pull of selection across multiple levels.

## C. Stasis as the Key to the Hierarchy

The phenomenon of stasis deserves special attention because it provides the conceptual lynchpin connecting punctuated equilibrium to multilevel selection. Gould identified multiple mechanisms that produce stasis: developmental constraints that channel variation into limited pathways, habitat tracking where species follow their preferred environment rather than adapting to new conditions, normalizing selection that eliminates deviants without producing directional change, and fluctuating directional selection that cancels out over geological timescales. The critical insight is that stasis is fully compatible with intense micro-level selection; a species can experience vigorous organismal competition without undergoing net directional change, just as a legal regime can absorb continuous judicial reinterpretation without structural transformation.

Species maintain their coherence and individuality not through the kind of physiological integration that holds organisms together, but through the continuous reproductive interaction of their constituent organisms via sexual reproduction and the evolution of reproductive isolating mechanisms that strictly exclude members of other species. This boundary maintenance gives species the discrete births (speciation), lifespans (stasis), and deaths (extinction) required for Darwinian individuality. They qualify as the 'atoms' of macroevolution, the irreducible units whose differential success generates large-scale evolutionary trends.

## D. Why the Debate Dissolves in Cultural Evolution

The Dawkins-Gould dispute is intractable in biology because the Weismann barrier forces a choice. Since acquired phenotypic modifications cannot enter the germline, the only entities with genuine replicator properties are genes. Gould's interactors at higher levels may be causally relevant, but their contributions must ultimately be transmitted through genes, creating an inherent asymmetry.

In cultural evolution, this asymmetry disappears. When a judge applies and modifies a precedent, the modification is simultaneously a somatic act (affecting the current case) and a germinal act (entering the hereditary stream of legal doctrine). This is what both Dawkins and Gould recognized as the Lamarckian character of cultural inheritance.

The absence of the Weismann barrier has a paradoxical consequence. One might expect that without a barrier separating phenotypic modification from hereditary transmission, the hierarchy would collapse: everything would be simultaneously replicator and interactor, and level distinctions would become meaningless. Legal systems, however, exhibit the opposite pattern. They display robust hierarchical structure with sharp differences in evolutionary tempo across levels. Doctrinal interpretation shifts within months. Legislative regimes persist for decades. Constitutional orders endure for centuries.

The explanation lies in the nature of the emergent properties that define each level. Constitutional lock-in is not a property of individual norms but of the architecture that connects them: the amendment rules, the judicial review scope, the treaty hierarchy, the distribution of veto powers. These architectural properties function as emergent fitness traits. They determine the persistence or extinction of entire normative orders through mechanisms that cannot be decomposed into the fitness of individual norms, just as geographic range determines the extinction risk of a species through mechanisms irreducible to the fitness of individual organisms.

Consider the analogy more carefully. A biological species maintains its individuality through reproductive isolation mechanisms and continuous gene flow among its constituent organisms. An individual organism may be perfectly healthy and reproductively successful, yet its species may go extinct if its geographic range shrinks below a critical threshold. The organism-level fitness and the species-level fitness are decoupled: the former depends on the organism's phenotypic match to its local environment, while the latter depends on emergent properties of the population as a whole. In legal evolution, the same decoupling operates. An individual norm may be doctrinal well-established and frequently cited, yet the normative regime it belongs to may collapse if the coalition supporting the regime fragments, if the regime's internal coherence is disrupted by contradictory legislative amendments, or if an exogenous shock displaces the system from its basin of attraction. The fitness of the individual norm and the fitness of the regime are related but not reducible to each other.

This point can be stated with greater precision using the distinction between what Gould called 'sorting' and 'selection.' Sorting is the differential survival or proliferation of entities at a given level; it may or may not result from selection at that level. When a species goes extinct because its constituent organisms are individually unfit, the species-level sorting is a byproduct of organism-level selection; Vrba calls this the 'effect hypothesis.' When a species goes extinct because its emergent properties (geographic range, population structure, speciation rate) make it vulnerable to macroevolutionary forces, the species-level sorting results from genuine species-level selection. In legal evolution, much of what appears to be macro-level change is actually effect-level sorting: individual norms fail because they are individually unfit, and the regime collapses as a byproduct. But some macro-level change results from genuine macro-level selection, operating on emergent properties of the constitutional architecture. The hierarchical model provides the theoretical framework for distinguishing these two scenarios, and the CLI provides the empirical tool for measuring the basin depth that determines which scenario predominates.

## III. A Three-Tier Hierarchical Model of Legal Evolution

## A. The Micro Level: Lamarckian Adaptation Through Heteronomous Bayesian Updating

At the finest grain of legal evolution, individual norms and doctrinal rules change through Heteronomous Bayesian Updating (HBU). Legal professionals do not learn the content of norms by studying their substantive purposes or evaluating their consequences; they learn by observing the reactions of institutional gatekeepers, principally appellate courts, to particular normative claims. A lawyer updates their understanding of what a rule means not by examining whether the rule achieves its stated objectives, but by observing whether invoking the rule in a particular way provokes reversal, affirmation, or indifference from the relevant tribunal.

HBU is Lamarckian in a precise sense. The modification acquired during the lifetime of a doctrinal application enters the hereditary stream immediately. The next lawyer citing that decision inherits not only the original rule but the modification. There is no separation between the somatic act of applying the rule and the germinal act of transmitting it. This makes doctrinal evolution extraordinarily rapid, but it also introduces a distinctive pathology: the content of norms drifts toward whatever institutional gatekeepers reward, which may or may not correlate with the norms' stated purposes or social utility.

The genealogy of HBU connects to a tradition older than its explicit formulation. John Austin's command theory of law, stripped of the Hartian mediation that domesticated it, already contained the essential insight: the sanction functions as an institutional signal that instantiates the norm in the observable world. Austin's sovereign is not a moral authority but a signal generator; the subjects learn what the law 'is' by observing what the sovereign punishes. HBU generalizes this mechanism beyond Austin's dyadic sovereign-subject model to a distributed network of institutional gatekeepers, each generating signals that legal practitioners aggregate and respond to through Bayesian updating.

The micro level corresponds to Gould's organismal level in biological evolution: it is where classical selection operates most directly, where the feedback loop between variation and differential replication is tightest, and where the bulk of observable adaptation occurs. Individual norms compete for citation, for inclusion in curricula, for adoption by practitioners. The fittest norms, in memetic terms, are those that replicate most successfully, not those that produce the best social outcomes. This is Dennett's free-floating rationale at the legal level: norms persist because they are good at persisting, and the reasons for their persistence may never enter the conscious awareness of the practitioners who transmit them.

The Lamarckian character of micro-level transmission has a further consequence that distinguishes legal evolution from both biological evolution and other forms of cultural evolution: the rate of mutation is not independent of the environment. In genetic evolution, mutation rates are essentially random with respect to adaptive need (the occasional exception of stress-induced mutagenesis notwithstanding). In legal evolution, the rate and direction of doctrinal mutation correlate directly with environmental pressures. When economic conditions deteriorate, courts tend to reinterpret contractual provisions more favorably to weaker parties; when regulatory pressure intensifies, compliance doctrine adapts rapidly. This directed mutation is possible only because of the absence of the Weismann barrier: the same actors who experience the environmental pressure are the actors who modify the hereditary material. Directed Lamarckian mutation at the micro level coexists with undirected stochastic change at the macro level, a combination that has no analogue in biological evolution.

This directed quality of micro-level mutation explains a puzzling feature of legal systems: their extraordinary capacity for local adaptation combined with their equally extraordinary resistance to global change. A legal system can adjust its doctrinal interpretation to accommodate a new economic reality within months, yet the same system may resist fundamental structural reform for decades. The Lamarckian mechanism makes local adaptation fast and responsive; the hierarchical structure makes global change slow and punctuated. The two patterns are not contradictory but complementary, and their coexistence is a direct prediction of the model.

## B. The Meso Level: Memeplex Competition and Emergent Fitness

Individual norms do not evolve in isolation. They form complexes: memeplexes that legal scholars recognize as doctrinal fields, statutory regimes, or branches of law. Argentine labor law is not a collection of independent norms; it is an integrated normative system where the employment stability regime, the collective bargaining framework, the judicial enforcement apparatus, and the constitutional protections form a self-reinforcing whole. Modifying any single element triggers compensatory responses from the others, a phenomenon captured by the concept of institutional hysteresis.

The meso level is where the analogy to Gould's species selection becomes most precise. Normative regimes exhibit a characteristic lifecycle: rapid origination (a burst of legislative and judicial activity that establishes the regime), prolonged stasis (the regime persists without fundamental structural change, even as its individual components undergo constant micro-level adaptation), and eventual extinction (the regime is replaced wholesale during a period of institutional crisis or political rupture). Gould's example of the evolution of books illuminates the pattern. The physical form of the written record has progressed through sharp punctuations, from clay tablet to parchment scroll to codex to electronic edition, separated by extended periods of morphological stasis. Gutenberg's printing press, a technological revolution of the first magnitude, did not alter the morphology of the codex; it was superimposed onto the existing phenotype without changing its structure. Normative regimes behave analogously: they absorb enormous micro-level variation without structural transformation, then undergo rapid replacement when the entire regime is displaced by a competitor.

The trends observed in legal evolution at this level, the expansion of human rights regimes, the global spread of commercial law harmonization, the convergence of corporate governance standards, result from the differential proliferation and extinction of normative regimes, not from the gradual transformation of individual norms. This is macroevolution in Gould's precise sense: a genuinely irreducible higher-level process driven by the differential success of stable entities.

The critical question is whether memeplex-level selection is genuinely irreducible to the micro level. Gould's criterion of emergent fitness provides the answer. A normative regime's persistence depends on properties that are not attributes of its individual norms: its internal coherence (the degree to which its components mutually reinforce each other), its institutional embeddedness (the extent to which dedicated enforcement machinery exists), and its coalition support (the alignment of powerful actors whose interests the regime serves). The Argentine labor law regime provides the clearest illustration: individual norms within it are, by many metrics, individually unfit (high compliance costs, widespread evasion in the informal sector, regularly identified as impediments to growth), yet the regime persists because its structural properties confer system-level fitness that overrides individual-norm unfitness.

The recent dialogue between Joseph Henrich and Robin Hanson on cultural evolution provides independent empirical corroboration for this meso-level mechanism. Henrich argues, on the basis of decades of cross-cultural research, that humans are fundamentally incapable of designing effective institutions from scratch. Institutional complexity routinely exceeds the cognitive capacity of any individual designer; successful institutions incorporate hidden structures, ritual elements, and social practices whose functional rationale is causally opaque to the participants themselves. This 'causal opacity' of institutional design means that institutions must evolve through variation and selection rather than rational engineering. Different communities, jurisdictions, and organizations function as local experiments, testing distinct normative configurations; the mechanism that sifts through these experiments is cultural group selection driven by intergroup competition.

Patrick Francois's study of U.S. banking deregulation between the 1960s and 1992 provides a concrete empirical illustration. As states progressively deregulated their banking sectors, increasing inter-firm competition, generalized trust among individuals in those states rose measurably. The mechanism is precisely what the hierarchical model predicts at the meso level: competitive pressure among normative regimes selects for configurations that promote internal cooperation, suppress free-riding, and build the social capital necessary for collective survival. Market competition functions as a domesticated form of intergroup selection, forcing institutions to evolve toward configurations that enhance internal cohesion. This is species selection applied to legal regimes: the differential survival and proliferation of normative systems based on their emergent fitness properties, not on the individual fitness of their component norms.

Henrich's observation connects to the historical record that Gould and others have identified. Periods of exceptional institutional innovation, the Greek city-states, the Italian Renaissance communes, the competing polities of early modern Europe, were characterized by high jurisdictional variation and intense intergroup competition. Each independent polity ran its own normative experiment; successful innovations were copied through prestige-biased transmission; unsuccessful experiments were eliminated through competitive failure. This pattern is punctuated equilibrium at the meso level: long periods of normative stasis within individual jurisdictions, interrupted by episodes of rapid institutional change driven by competitive pressure from more successful neighbors.

Comparative analysis across jurisdictions strengthens the emergent fitness claim. Brazil's labor law regime (CLI = 0.40) exhibits a different pattern. The 2017 Temer reform succeeded in modifying significant portions of the Consolidation of Labor Laws (CLT), a regime that had persisted since 1943. The reform's partial success was possible because Brazil's normative regime, while structurally coherent, lacked the extreme institutional embeddedness of its Argentine counterpart: Brazilian labor courts are integrated into the ordinary judiciary rather than constituting a separate judicial branch, and the union structure allows for greater plurality than Argentina's monopolistic model. The meso-level emergent fitness of the Brazilian regime was lower than the Argentine equivalent, permitting reform displacement that would be impossible in the Argentine basin. Spain's labor market reforms under the 2012 Rajoy government offer another instructive comparison. Spain (CLI = 0.51) achieved structural modification of its labor regime, including the relaxation of collective bargaining supremacy and the reduction of dismissal costs, but these reforms required the extraordinary political capital generated by the deepest economic crisis in Spanish post-transition history. The reforms have partially persisted through subsequent changes in government, suggesting that Spain's meso-level basin, while shallower than Argentina's, is deep enough to partially re-absorb perturbation once the crisis-generated political energy dissipates.

These comparative cases illustrate a gradient of emergent fitness across jurisdictions. The meso-level fitness of a normative regime is not binary (fit or unfit) but continuous, and it correlates, as the model predicts, with the macro-level CLI that measures the depth of the constitutional basin of attraction. The correlation is not perfect, because meso-level fitness depends on structural properties that are partially independent of the formal constitutional architecture (coalition support, for instance, is a political rather than constitutional variable). But the systematic relationship between CLI and regime persistence, confirmed across all four jurisdictions in the dataset, provides strong preliminary evidence for the emergent fitness prediction.

## C. The Macro Level: Constitutional Lock-in and Punctuated Equilibria

At the broadest scale, legal evolution follows the punctuated equilibrium pattern: long periods of stasis interrupted by brief episodes of rapid, fundamental change. Constitutional orders are the 'species' of this macro-level evolutionary process. They originate in concentrated episodes of institutional design, persist for extended periods with remarkable structural stability, and terminate through discrete events rather than gradual transformation.

The Constitutional Lock-in Index (CLI) provides a quantitative measure of basin depth. CLI integrates four dimensions: textual vagueness of constitutional provisions (weighted at 40%), the position of international treaties in the normative hierarchy (30%), the intensity of judicial activism (20%), and the binding force of judicial precedent (10%). A regression model applied to 60 reform episodes across four jurisdictions produces an R-squared of 0.74: each 0.1 increase in CLI reduces reform success odds by 63% (OR = 0.37, 95% CI [0.21-0.64], $p < 0.001$).

The macro level exhibits what Gould emphasized for biological macroevolution: it is not fractal with respect to the micro level. The forces that dominate at the macro level, veto player configurations, supermajority requirements, judicial review scope, are qualitatively different from those driving micro-level doctrinal change. The coexistence of rapid micro-level Lamarckian adaptation with rigid macro-level stasis is a direct consequence of the hierarchy: each level has its own dominant mechanisms and characteristic tempo.

The mechanisms that produce constitutional stasis parallel those Gould identified for biological stasis with remarkable precision. Developmental constraints in biology channel variation into limited pathways; in constitutional law, amendment procedures and judicial review doctrines channel reform proposals into narrow channels that the existing architecture can absorb. Habitat tracking in biology occurs when species follow their preferred environment rather than adapting to new conditions; in constitutional law, institutional actors relocate their activities within the existing normative framework rather than seeking to change it. Normalizing selection eliminates biological deviants without producing directional change; constitutional review eliminates unconstitutional innovations without altering the constitutional order. And fluctuating directional selection, which cancels out over geological timescales, finds its analogue in the oscillation of political control between parties with different reform agendas, producing alternating micro-level shifts that cancel out at the macro level.

The critical empirical finding is that macroeconomic crises intensify rather than relax constitutional lock-in. The conventional political economy narrative treats crises as 'windows of opportunity' for structural reform, but the data from 60 reform episodes tell a different story. During crisis periods, judicial invalidation rates increase significantly, indicating that macro-level constitutional architecture activates homeostatic defense mechanisms that resist displacement. This is analogous to the biological observation that environmental stress often increases developmental canalization (the tendency for development to produce the same phenotype despite environmental perturbation) rather than decreasing it. The constitutional order, like the species phenotype, becomes more resistant to perturbation precisely when perturbation is most intense.

The Argentine case illustrates the point with exceptional clarity. Twenty-three reform attempts have failed since 1991, producing a zero success rate at the macro level. Yet judicial interpretation of labor provisions has shifted continuously, with courts adjusting doctrinal applications in response to changing economic conditions, international human rights jurisprudence, and shifting political equilibria. The median time to reversal for the few reforms that achieved initial implementation is 18.3 months. The behavioral plasticity at the micro level coexists with absolute structural rigidity at the macro level. The empirical data from crisis periods further sharpen the picture: the interaction coefficient between CLI and crisis dummy variables is negative (Beta = -2.83, p = 0.014), meaning that macroeconomic crises intensify rather than relax constitutional lock-in, contradicting conventional political economy that treats crises as windows of opportunity.

## D. Cross-Level Interactions: Upward and Downward Causation

The three tiers do not operate in isolation. They interact through mechanisms that Gould characterized as upward and downward causation. Upward causation, which Vrba termed the 'effect hypothesis,' occurs when micro-level adaptations cascade to produce meso-level consequences that no individual actor intended. A sequence of individually motivated judicial decisions on consumer protection, each responding to the specific facts of a particular case, can accumulate to produce a new doctrinal regime, a meso-level entity, without any court having designed it. The regime emerges as a byproduct of micro-level selection, not as the product of deliberate institutional engineering.

Downward causation occurs when macro-level structures constrain the space of micro-level variation. In Argentina, the constitutionalization of labor protections (Article 14 bis CN) exercises downward causation on judicial interpretation by filtering out doctrinal innovations that would weaken the structural labor protection framework, regardless of their economic merit. The constitutional architecture acts as a channel constraint in Gould's sense: it does not determine the direction of micro-level change, but it restricts the permissible range of variation. This is analogous to how developmental constraints in biology do not dictate which mutations occur, but channel the phenotypic expression of mutations into limited pathways.

Cross-level interactions can be synergistic, orthogonal, or antagonistic. The Argentine labor case exemplifies antagonistic interaction: micro-level adaptations that would move the system toward greater labor market flexibility are consistently blocked by macro-level constitutional constraints. The Chilean case, by contrast, exhibits more synergistic interaction: micro-level doctrinal innovations propagate more readily to the meso and macro levels because the shallow basins of attraction (CLI = 0.24) offer less resistance to upward causation. Understanding the predominant mode of cross-level interaction in a given system is essential for predicting reform outcomes.

The distinction between sorting and selection, introduced by Gould to clarify the logic of multilevel analysis, applies directly to cross-level causation in law. Not every macro-level outcome results from genuine macro-level selection. When a normative regime collapses because its individual norms are so dysfunctional that practitioners abandon them one by one, the regime-level extinction is an effect of micro-level selection, not a product of regime-level dynamics. When a normative regime persists despite the individual unfitness of its component norms, maintained by emergent structural properties that resist displacement, the regime-level persistence is the product of genuine meso-level selection operating on emergent fitness. The empirical challenge is to distinguish these two scenarios, and the CLI provides the critical diagnostic: in high-CLI systems, the predominance of meso and macro-level selection over micro-level effects is predicted to be stronger, because the deep basins of attraction insulate higher-level entities from the accumulation of micro-level perturbations.

An analogy from cell biology may clarify the mechanism. In a healthy multicellular organism, cell-level competition is actively suppressed by organism-level integration mechanisms: the immune system, apoptosis (programmed cell death), and contact inhibition prevent rogue cells from proliferating at the expense of the organism. When these suppression mechanisms fail, the result is cancer: cell-level selection reasserts itself against organismal interests. In legal systems, the constitutional architecture functions analogously to organism-level integration mechanisms: it suppresses micro-level variations that would disrupt macro-level structural integrity. When a micro-level doctrinal innovation threatens constitutional coherence, judicial review eliminates it, just as the immune system eliminates mutant cells. When the suppression mechanism fails, as occurs during periods of constitutional crisis or when the judiciary lacks independence, micro-level selection reasserts itself and the constitutional order may fragment, the institutional equivalent of cancer. This biological analogy is more than decorative; it captures a genuine structural parallel between the mechanisms that maintain coherence across levels in biological and legal hierarchies.

## IV. Are Legal Norms a Paradigm Darwinian Population? The Godfrey-Smith Challenge

Peter Godfrey-Smith's framework for evaluating Darwinian populations poses a serious challenge to any memetic theory of legal evolution. Godfrey-Smith distinguishes paradigm Darwinian populations, those capable of generating novel, complex, and highly adapted structures, from marginal cases that only loosely approximate the minimal criteria for evolution by natural selection. The distinction turns on several parameters: the fidelity of heredity (H), the abundance and character of variation (V), the smoothness of the fitness landscape (C), the dependence of fitness on intrinsic characteristics (S), and the degree to which reproduction involves bottlenecks and germ-soma differentiation.

Godfrey-Smith is skeptical of memes as paradigm Darwinian populations. If cultural learners are 'too smart,' meaning they blend multiple models, conform to majorities, or customize behaviors through intelligent synthesis, the clear parent-offspring lineages required for cumulative Darwinian evolution dissolve. The result is not paradigm evolution but marginal, partially Darwinian change. This critique strikes at a vulnerability in naive memetics: the assumption that cultural units replicate with gene-like fidelity.

Legal norms, however, occupy a distinctive position in Godfrey-Smith's framework that other cultural replicators do not. Three features push legal norm populations toward the paradigm end of the spectrum. First, legal norms possess unusually high hereditary fidelity (H). Written legal texts, unlike oral traditions, are copied with near-perfect accuracy across generations. A statute enacted in 1853 is transmitted to present-day practitioners in essentially the same form. Latin legal maxims, as I have argued elsewhere, achieve perfect copying-fidelity across millennia precisely because they are encoded in a dead language immune to semantic mutation. The parameter H, for written legal norms, approaches values comparable to genetic replication.

Second, legal norm populations exhibit clear reproductive structure with something approaching a germ-soma distinction. Not all legal practitioners contribute equally to the hereditary stream. Appellate judges, legislators, and influential legal scholars function as a 'germline' whose outputs enter the canon and are transmitted to future generations. The vast majority of legal practitioners function as 'soma': they apply norms but do not modify the hereditary material. This reproductive specialization, while not as rigid as biological germ-soma separation, gives legal norm populations a structure that oral cultural traditions lack.

Third, the fitness landscape for legal norms exhibits moderate continuity (C). Small modifications in legal doctrine typically produce small changes in institutional response, not catastrophic discontinuities. This smooth landscape permits the incremental, cumulative evolution that Godfrey-Smith requires for paradigm status. The smoothness is not universal; constitutional amendments involve discontinuous fitness shifts. But within the micro level of doctrinal evolution, the landscape is smooth enough to support sustained cumulative adaptation.

The Godfrey-Smith challenge thus does not refute the memetic approach to law; it refines it. Legal norm populations are not merely marginal Darwinian populations; they are among the most paradigm-like cultural populations, precisely because written transmission, institutional reproductive specialization, and landscape continuity push them toward the conditions that make cumulative Darwinian evolution possible. This refinement strengthens rather than weakens the hierarchical model, because it supports the claim that the micro-level Darwinian process is genuine and robust, providing the foundation on which the meso and macro levels are built.

Godfrey-Smith's warning against 'Darwinian paranoia,' the temptation to attribute agential, purposeful behavior to mindless population dynamics, deserves careful attention. The language of memetics is particularly susceptible to this error: speaking of 'selfish' norms that 'compete' for 'survival' can create the impression that norms are strategic agents. The hierarchical model proposed here avoids this trap by using the language of selection and fitness in a strictly population-dynamic sense. When I say that a norm is 'fit,' I mean that it exhibits high replication rates relative to alternatives, not that it possesses any kind of strategic intelligence. When I say that normative regimes 'compete,' I mean that regimes with higher emergent fitness proliferate at the expense of those with lower emergent fitness, not that regimes are intentional agents pursuing goals. The anthropomorphic language is a heuristic convenience, not an ontological commitment.

A further point from Godfrey-Smith's framework merits discussion. His parameter S measures the degree to which fitness depends on intrinsic characteristics of the entity versus its extrinsic environment. In paradigm biological populations, S is moderately high: an organism's fitness depends substantially on its own phenotype. In legal norm populations, S is remarkably low: a norm's replication success depends overwhelmingly on its institutional environment, its position within a normative regime, and the political coalition supporting it. The same consumer protection provision might be highly fit in an environment with an active regulatory agency and accessible courts, and completely inert in an environment lacking enforcement infrastructure. This low S value is precisely why the IusSpace framework is essential: it maps the environmental topology that determines which norms will thrive and which will fail, making the extrinsic determinants of fitness explicit and measurable.

## V. IusSpace as Environmental Constraint Topology

Biological evolution operates within a fitness landscape: a mapping from phenotype space to fitness values that determines which evolutionary trajectories are possible and which configurations are forbidden. Legal evolution requires an analogous concept, and IusSpace provides it.

IusSpace is a twelve-dimensional mathematical space in which every possible legal system configuration occupies a unique point, defined by twelve fundamental institutional dimensions: separation of powers, federalism, individual rights, judicial review, executive power, legislative scope, amendment flexibility, interstate commerce, constitutional supremacy, emergency powers, social rights, and international integration. Each dimension ranges from 1 to 10, producing 10^12 possible configurations. Legal reforms are vectors through this space; their success depends on the distance traveled, the direction of movement, and the properties of the region traversed.

The relevance of IusSpace to the hierarchical model is fourfold. First, IusSpace defines the environmental constraints within which selection operates at all three levels. A norm's fitness is not intrinsic; it depends on position within a specific institutional environment. The same norm thriving in a high-judicial-review system may fail where courts lack enforcement power. This is what Godfrey-Smith captures with his parameter S: the degree to which fitness depends on intrinsic versus extrinsic characteristics. In legal evolution, S is low; fitness is overwhelmingly environment-dependent.

Second, IusSpace maps the structure of the basins of attraction that generate macro-level stasis. CLI can be understood as a curvature property of IusSpace: systems with high CLI values occupy deeply concave regions, creating strong restoring forces whenever reform displaces them. Systems with low CLI values occupy flatter regions where displacement encounters less resistance.

Third, IusSpace makes visible the WEIRD/non-WEIRD bifurcation that Henrich's framework predicts. Empirical mapping of eighteen reforms across thirteen countries revealed that WEIRD societies exhibit implementation gaps averaging 5.4%, while non-WEIRD societies show gaps averaging 31.2% ($p < 0.0001$, Cohen's $d = 3.749$). This bifurcation reflects fundamentally different relationships between formal norms and informal institutional constraints.

The connection to Henrich's gene-culture coevolution is direct. WEIRD societies, shaped by centuries of Catholic Church-driven dissolution of intensive kinship institutions, evolved psychological profiles characterized by individualism, analytical thinking, and reliance on impersonal rules. These profiles create environments where formal norms and behavior track each other closely. Non-WEIRD societies, where kinship institutions remain dominant, exhibit systematic gaps between formal law and practice. The norm psychology that Henrich identifies, the human tendency to internalize rules as ends in themselves, to punish violators, and to calibrate behavior to prestige and conformity biases, operates differently across these two institutional environments. In WEIRD contexts, the relevant prestige models are formal institutional actors (judges, legislators, regulators) whose signals propagate through official channels. In non-WEIRD contexts, prestige hierarchies rooted in kinship, religious authority, and local community standing compete with and often override formal institutional signals.

Henrich's concept of self-domestication adds a deeper evolutionary layer. Over millennia, living in norm-governed environments created selective pressures favoring docile, rule-following, and socially sensitive individuals. Communities punished norm violators through gossip, reputation damage, ostracism, and in extreme cases, execution. This persistent cultural selection on human genetics produced a species that is, in Henrich's phrase, 'wired for culture': predisposed to internalize norms, monitor others' compliance, and derive satisfaction from punishing defectors. The self-domestication process operated with different intensities across institutional environments. Populations exposed to intensive kinship institutions developed norm psychologies oriented toward in-group loyalty, conformity to elder authority, and shame avoidance. Populations exposed to post-kinship market-based institutions developed norm psychologies oriented toward impersonal fairness, individual achievement, and guilt avoidance. These distinct psychological profiles produce the different relationships between formal law and actual behavior that IusSpace captures as the WEIRD/non-WEIRD implementation gap.

The hierarchical model proposed in this Article interacts with Henrich's framework in a specific way. The three tiers of legal evolution operate in both WEIRD and non-WEIRD environments, but the coupling between tiers differs systematically. In WEIRD societies, self-domestication has produced psychological profiles that are highly responsive to formal institutional signals; the HBU mechanism at the micro level transmits changes that propagate relatively freely to the meso and macro levels. In non-WEIRD societies, self-domestication has produced psychological profiles that are responsive to kinship-based rather than formal institutional signals; the HBU mechanism operates within the formal legal system but its outputs are modulated, filtered, and sometimes nullified by the parallel informal institutional substrate. This differential coupling explains why the same CLI value may predict different reform outcomes in WEIRD versus non-WEIRD contexts: the formal architecture is only one determinant of basin depth, and in non-WEIRD systems, the informal architecture contributes a 'dark matter' of institutional resistance that CLI's four formal dimensions do not capture.

Fourth, and most importantly for the present argument, IusSpace provides the topology within which cross-level interactions operate. Upward causation (micro-level innovations cascading to produce meso-level regime change) is constrained by the curvature of IusSpace at the system's current position. In deeply concave regions (high CLI), upward causation is damped: micro-level innovations are absorbed without producing macro-level displacement. In flatter regions (low CLI), upward causation propagates more freely. This topological perspective transforms the abstract notion of 'cross-level interaction' into a geometrically precise concept amenable to measurement and prediction.

The relationship between IusSpace and Dennett's Design Space deserves explicit articulation. Dennett conceives of all possible biological designs as occupying positions in a vast, multidimensional space, with evolution tracing paths through this space under the guidance of natural selection. IusSpace is a legal Design Space: all possible legal system configurations occupy positions in a twelve-dimensional continuum, with legal evolution tracing paths under the guidance of institutional selection. The inhabited regions of IusSpace, like the inhabited regions of biological Design Space, represent configurations that have proven viable under actual historical conditions. The uninhabited regions represent configurations that are either institutionally contradictory (a system with maximum executive power and maximum separation of powers simultaneously) or socially rejected (configurations that no society has found compatible with its cultural substrate). Dennett's insight that Design Space contains far more empty space than inhabited space applies with full force: the overwhelming majority of the 10^12 possible IusSpace configurations have never been realized and probably never will be. Legal evolution, like biological evolution, explores only a tiny fraction of the available design possibilities, constrained by path dependence, historical contingency, and the structural requirements of institutional viability.

## VI. Bunge's Systemist Critique and the Strengthening of Extended Phenotype Theory

Mario Bunge's philosophical framework presents what appears to be a fundamental challenge to the Extended Phenotype approach. Bunge characterized Dawkins' gene-centered view as an extreme form of biological determinism, arguing that it reduces organisms to 'tubes for transmitting genes' and that natural selection operates on whole organisms and populations, not on isolated genes. More broadly, Bunge's emergentist materialism holds that reality is organized into genuine levels, each exhibiting emergent properties that require level-specific concepts.

Bunge's CESM model (Composition, Environment, Structure, Mechanism) offers rigorous criteria for identifying genuine levels of organization. A system is real if it possesses a composition of interacting parts, an environment it interacts with, a structure of internal and external relationships, and mechanisms that produce its characteristic behaviors. Emergent properties arise from the Structure, not from the Composition alone. A bag of molecules is not an organism; the organism's emergent properties, life, metabolism, reproduction, arise from the structural relationships among its molecular components, not from the molecules considered individually.

This critique is well-taken when directed at a naive interpretation of EPT. No serious version of the framework treats norms as if they were literal genes whose fitness can be calculated independently of context. The norm-as-replicator analogy identifies the unit of cultural inheritance, not the unit of causal interaction. The institution-as-extended-phenotype concept recognizes that the material expressions of normative competition are complex systems with emergent properties.

What Bunge's systemism adds is the philosophical infrastructure needed to make the hierarchical model rigorous. Applied to a normative regime, the CESM model specifies: Composition (individual norms and doctrinal rules), Environment (the political system, economy, cultural context), Structure (logical relationships among norms, institutional hierarchies, professional networks), and Mechanism (HBU, legislative revision, judicial enforcement). The emergent fitness of the regime depends on Structure and Mechanism, not merely Composition. This is why regimes composed of individually fit norms can fail as systems, and why regimes containing individually suboptimal norms can persist if their structural properties confer system-level advantages.

Bunge's distinction between ontological reduction and epistemological reduction is particularly illuminating. The claim that legal norms are memes is an ontological claim about the nature of legal evolution. It does not entail the epistemological claim that legal phenomena can be fully explained at the level of individual norms. Just as the ontological reduction of mental processes to brain processes does not justify reducing psychology to neurophysiology (because neural processes are influenced by social circumstances requiring psychology's own concepts), the ontological identification of norms as replicators does not justify reducing institutional dynamics to norm-level analysis. The meso and macro levels require their own explanatory frameworks, with their own concepts (memeplex coherence, institutional hysteresis, constitutional lock-in), precisely because structural relationships generate emergent properties individual norms do not possess.

Bunge would object to calling norms ‘selfish,’ and rightly so: selfishness requires intentionality, which norms lack. But the metaphor was always a heuristic for differential replication, not an attribution of literal intentionality. As discussed in Section IV, Godfrey-Smith’s caution against ‘Darwinian paranoia’ applies here with equal force. The hierarchical model treats norms as replicators in a technical sense, entities exhibiting heritable variation in fitness, rather than as agents with strategies or goals.

Bunge's advocacy of moderate reductionism, as opposed to both radical reductionism and holism, maps directly onto the hierarchical model's structure. Moderate reductionism acknowledges that higher-level systems are ontologically rooted in lower-level components while insisting that emergent properties at each level require level-specific concepts and methods for their explanation. The hierarchical model is moderately reductionist in precisely this sense: it recognizes that meso-level normative regimes are composed of micro-level norms (ontological reduction) while insisting that regime-level dynamics require concepts like emergent fitness, institutional hysteresis, and coalition support that have no meaning at the individual-norm level (epistemological irreducibility). Similarly, macro-level constitutional orders are composed of meso-level normative regimes, but their persistence depends on basin depth, amendment architecture, and veto player configurations, properties that exist only at the system level.

Bunge's insistence that authentic scientific explanation requires identification of concrete mechanisms rather than mere description of regularities reinforces the model's emphasis on specifying the mechanisms operating at each level. The micro-level mechanism is HBU: legal professionals observe institutional reactions and update their doctrinal priors accordingly. The meso-level mechanisms include legislative bargaining, judicial doctrine building, and institutional embeddedness through dedicated enforcement agencies. The macro-level mechanisms include constitutional amendment procedures, judicial review doctrines, and the veto player configurations that determine how much political energy is required to displace the system from its basin. Each mechanism is concrete, identifiable, and empirically investigable, meeting Bunge's standard for scientific explanation rather than relying on abstract 'fitness' talk disconnected from causal processes.

## VII. Exaptation, Contingency, and Institutional Repurposing

Gould and Vrba's concept of exaptation, the co-option of a pre-existing structure for a new utility distinct from the function for which it originally evolved, illuminates a pervasive but undertheorized phenomenon in legal evolution: institutional repurposing. The distinction between historical genesis and current utility, which Gould traced to Nietzsche, applies with full force to legal institutions. The reason a legal structure was created is often entirely disconnected from the function it performs today.

The Argentine collective bargaining framework offers a concrete example. Designed in the mid-twentieth century to organize labor relations in an industrial economy, it has been exapted as a political coordination mechanism for distributing state resources to allied constituencies. The structure persists not because it performs its original function well (industrial labor relations have declined as the economy de-industrialized) but because it has acquired a new function (political resource distribution) for which it is well-adapted. This is not a dysfunction in the evolutionary sense; it is a predictable feature of hierarchical evolution. Structures that persist at the meso level develop new functional relationships with their environment, acquiring aptations through co-option rather than through direct selection for the new function.

Pievani's analysis of evolutionary contingency deepens this insight. Evolution, he argues, is not a march toward optimization but an intricate labyrinth of possibilities shaped by historical accident and path dependence. The 'bricolage' metaphor, borrowed from Francois Jacob, captures how natural selection works with available materials rather than designing from scratch. The same applies to legal evolution: institutional reformers cannot design legal systems de novo but must work with the existing institutional materials, reshaping and repurposing structures inherited from prior evolutionary episodes.

The concept of the 'exaptive pool' is particularly relevant. Gould argued that non-adaptive spandrels, structural byproducts of other evolutionary developments, provide a massive reservoir of raw material for future co-option, guaranteeing evolutionary flexibility. Legal systems are rich in normative spandrels: provisions that were enacted for one purpose but that lie dormant until a new environmental challenge makes them useful for something entirely different. Argentina's Articles 1757-1758 of the Civil and Commercial Code, originally designed to govern objective liability in tort law, may prove to be an institutional spandrel available for exaptation as the normative foundation for pre-deployment evaluation of legislation, an application that the original drafters could not have anticipated.

The contingency perspective also challenges any teleological interpretation of legal evolution. If institutional outcomes depend on historical accidents, path dependencies, and the repurposing of structures for functions they were never designed to serve, then the history of law cannot be read as a narrative of progress toward optimal institutional arrangements. The Argentine labor law framework is not 'stuck' in a suboptimal equilibrium that rational reform could dislodge; it is a contingent outcome of a specific evolutionary trajectory that has produced a self-reinforcing system whose persistence depends on emergent properties, not on the optimality of its components. This is precisely analogous to Gould's argument that the history of life is not a tale of progressive improvement but of contingent, multilevel proliferation.

The analogy between biological and legal exaptation extends further. In biology, exaptation is intimately linked to what Gould and Lewontin called spandrels: structural byproducts of architectural constraints that were not built by selection for any function but that subsequently become available for co-option. Legal systems abound in normative spandrels. Consider the Argentine amparo proceeding, originally designed as a narrow remedy against arbitrary state action. Its vague constitutional formulation (Article 43 CN) was a spandrel of the 1994 constitutional reform, included without detailed specification because the political negotiation focused on other provisions. Over the following decades, courts exapted the amparo into a general-purpose rapid remedy for virtually any constitutional claim, including consumer protection, healthcare access, and environmental enforcement. The current function of the amparo bears almost no resemblance to its original design. Its textual vagueness, a structural byproduct of political negotiation, became the exaptive pool that enabled its transformation into one of the most powerful procedural tools in Argentine law.

A second example reinforces the point. The concept of 'abuse of right' (Article 10 CCyCN) was codified as a general corrective to the absolutist conception of subjective rights, a relatively modest doctrinal innovation. Through judicial co-option, it has been exapted into a flexible tool for adjusting contractual equilibria, controlling corporate behavior, and even modulating the exercise of fundamental rights. The historical genesis (a doctrine against absolute rights) is disconnected from the current utility (a general-purpose judicial adjustment mechanism). This is exaptation in its purest form: a structure evolved for one function is repurposed for an entirely different one, and the new function may become more evolutionarily important than the original.

Pievani's emphasis on the role of imperfection in evolutionary creativity deserves application here. Biological organisms, Pievani argues, are not optimally designed; they are 'bricolage' constructions assembled from historically available materials, full of compromises and suboptimal features. Yet these very imperfections provide the raw material for future innovation. The same principle applies to legal systems. The imperfections of legal architecture, its vague provisions, its contradictory mandates, its procedural gaps, are not merely failures of design; they constitute an exaptive pool that gives the system evolutionary flexibility. A perfectly specified legal system, one in which every provision had a single, unambiguous function, would be evolutionarily rigid: there would be no normative spandrels available for co-option when new environmental challenges arise. The 'messiness' of real legal systems, so frustrating to rationalist reformers, may be a feature rather than a bug from an evolutionary perspective: it provides the variation and redundancy that enable long-term adaptation.

The Henrich-Hanson dialogue raises a further alarm: the global homogenization of legal and institutional cultures threatens to deplete this exaptive pool. Hanson identifies four parameters that indicate institutional fragility: the reduction of cultural variety as thousands of autonomous normative traditions collapse into a single globalized model; the weakening of selection pressures as inefficient institutions persist through political subsidies rather than being eliminated by competitive failure; the acceleration of environmental change as economic complexity doubles every twenty years, outpacing the capacity of norms to stabilize; and the dominance of conformity pressures over functional adaptation. The transition from a world of high jurisdictional variation to a world of institutional monoculture is, in the evolutionary framework proposed here, a catastrophic reduction in the available pool of normative variation on which meso-level selection can operate. If all jurisdictions converge on the same normative regime, the differential survival and proliferation of competing regimes, the engine of meso-level evolution, stalls.

This diagnosis has a direct connection to the hierarchical model. A world with high jurisdictional variation is a world where meso-level evolution operates vigorously: competing normative regimes generate the variation on which cultural group selection can act, and the fittest regimes proliferate. A world of institutional monoculture is a world where meso-level evolution is suspended: there is no variation to select among, and the single surviving regime persists not because it is fit but because there are no competitors. In Gould's terms, this is analogous to a biosphere reduced to a single species: catastrophically vulnerable to environmental change because the diversity that provides adaptive resilience has been eliminated. The proposal for charter cities and jurisdictional experimentation, which Hanson and Henrich both endorse, can be understood within the hierarchical model as an attempt to restore meso-level variation and reactivate the engine of institutional evolution.

Gould extended the analogy of punctuated equilibrium beyond biology to the history of human artifacts, technology, learning theory, and the dynamics of organizations. His example of the evolution of the book, clay tablet to scroll to codex to electronic edition, with Gutenberg's press superimposed on the unaltered codex phenotype, captures the principle that technological revolutions do not necessarily alter the structural form of the artifact they transform. Baumgartner and Jones adapted the punctuated equilibrium concept to policy dynamics, showing that public policy exhibits long periods of incremental adjustment interrupted by sudden, dramatic shifts. Their empirical work on U.S. federal budgets demonstrated that budgetary allocations follow a leptokurtic distribution, exhibiting far more extreme changes and far more periods of near-zero change than a normal distribution would predict, precisely the statistical signature of punctuated equilibrium applied to institutional data.

The present Article extends this tradition to legal systems as a whole, arguing that the three-tier hierarchical model provides the theoretical infrastructure needed to explain why punctuated patterns recur across such diverse domains: the fundamental mechanisms of stasis (emergent fitness, basin depth, self-reinforcing structural properties) and punctuation (exogenous shock, coalition rupture, threshold displacement) are general features of hierarchical evolutionary systems, whether biological, cultural, or institutional. Lipton's application of evolutionary concepts to the history of company law provides a further bridge: he demonstrated that corporate law exhibits long periods of stability punctuated by episodes of rapid change driven by economic crises, and that path dependence and historical contingency, not rational design, explain the diversity of corporate governance regimes across jurisdictions. The convergence of evidence from paleobiology, policy studies, legal history, and the present four-jurisdiction comparison suggests that the hierarchical model identifies a genuinely general pattern, not a domain-specific artifact.

## VIII. Falsifiable Predictions and Preliminary Evidence

### A. The Decoupling Prediction

The hierarchical model predicts that micro-level Lamarckian plasticity should coexist with macro-level constitutional rigidity in high-CLI systems, and that the degree of decoupling should correlate positively with CLI values.

| Jurisdiction | CLI | Macro Reform Success | Micro Doctrinal Shift | Decoupling Index | Predicted by Model |
|---|---|---|---|---|---|
| Argentina | 0.89 | 0/23 (0%) | High (continuous) | Maximum | Yes |
| Spain | 0.51 | Partial | Moderate | Moderate | Yes |

| | | | | | |
|---|---|---|---|---|---|
| Brazil | 0.40 | Mixed | Moderate-High | Moderate | Yes |
| Chile | 0.24 | Bidirectional | High | Low | Yes |

*Table 1. Decoupling between micro-level adaptation and macro-level reform across four jurisdictions. CLI values are theoretically motivated estimations. Decoupling index represents the qualitative gap between observed micro-level plasticity and macro-level reform success.*

Argentina (CLI = 0.89) exhibits the most extreme decoupling: zero structural reform success combined with continuous doctrinal evolution. Chile (CLI = 0.24) shows tight coupling: constitutional amendments in both directions with relative ease, and micro-level changes propagating readily to the macro level. Brazil and Spain occupy intermediate positions consistent with their intermediate CLI values. The ordinal ranking is perfect: decoupling correlates monotonically with CLI across all four cases.

## B. The Emergent Fitness Prediction

The model predicts that memeplex-level outcomes should depend on system-level properties irreducible to the fitness of individual norms. The persistence of a normative regime should correlate more strongly with its structural properties than with the individual fitness of its component norms.

The Argentine labor law regime provides the strongest evidence. Individual norms within the regime are, by standard metrics, individually unfit: high compliance costs, widespread evasion (approximately 35-40% of the labor force operates informally), and they are regularly identified as impediments to investment and growth. Yet the regime persists because its structural properties, the self-reinforcing links between constitutional protections (Article 14 bis CN), judicial enforcement (specialized labor courts with pro-worker procedural rules), union organization (monopolistic representation per activity sector), and political coalition dynamics (organized labor as pivotal electoral constituency), confer system-level fitness that overrides individual-norm unfitness. An empirical analysis within the 'Evolución Constitucional' research notebook indicates that MLS2-type predictions (disfunctional norms imposing high social costs should collapse under intergroup competition) achieve 0% accuracy in the Argentine sample, while EPT predictions (norms persist based on memetic fitness through internal coalition support) achieve 100% accuracy.

## C. The Cross-Level Causation Prediction

The model predicts characteristic patterns of upward and downward causation detectable through citation network analysis and reform outcome data. Upward causation should be observable when micro-level innovations cascade to produce meso-level regime changes; downward causation should be observable when macro-level constraints shape micro-level evolution.

Genealogical analysis of 120 constitutional clauses across the four jurisdictions indicates that 91% of normative content possesses direct ancestors more than 50 years prior to supposed legal 'revolutions,' consistent with the stasis prediction. Architectural innovations, by contrast, exhibit a punctuated change rate of 52%, consistent with the macro-level punctuated equilibrium prediction. Downward causation is empirically detectable: during crisis periods, judicial invalidation rates increase significantly (OR = 2.83, 95% CI [1.42-5.67], $p = 0.003$), indicating that macro-level constitutional architecture activates homeostatic defense mechanisms that resist displacement.

The genealogical evidence merits closer examination. The RootFinder methodology, developed within this research program, traces the ancestral lineage of normative content by identifying textual, doctrinal, and functional predecessors across constitutional generations. Applied to Argentina's 1994 constitutional reform, widely considered a moment of dramatic institutional change, the analysis suggests that the reform was overwhelmingly a recombination of pre-existing normative material rather than a genuine origination event. The constitutional text adopted in 1994 drew 91% of its normative substance from sources that predated the reform by more than 50 years, and 67% from sources that predated it by more than 100 years. What appeared to contemporary observers as a constitutional revolution was, in evolutionary terms, more akin to lateral gene transfer or recombination: existing normative material was rearranged into a new configuration without the introduction of genuinely novel content.

The 52% rate of architectural innovation observed in the sample refers specifically to meta-norms: provisions governing amendment procedures, judicial review scope, federal structure, and rights hierarchies. These architectural changes are the 'speciation events' of constitutional evolution: they alter the structural relationships that generate emergent fitness, potentially creating new basins of attraction or modifying the depth of existing ones. The contrast between 91% content stasis and 52% architectural innovation captures the hierarchical structure in quantitative form: the norms themselves are mostly inherited (micro-level continuity), but the architecture connecting them changes more frequently (macro-level punctuation). This is precisely analogous to the biological observation that most genetic material is highly conserved across species while the regulatory architecture that controls gene expression evolves more rapidly.

A methodological note on the empirical evidence presented in this section: the CLI values used throughout this Article are theoretically motivated estimations derived from comparative institutional analysis, not the products of a fully validated empirical instrument. The four-jurisdiction comparison provides preliminary support for the model's predictions, but the small sample size ($n = 4$ for the CLI comparison, $n = 60$ for the reform episodes) limits the statistical power of the analysis. Expanding the sample to 20 or more jurisdictions, a project currently in development, will be necessary to establish robust statistical relationships and to control for confounding variables such as GDP per capita, legal tradition, and longitudinal political stability. The evidence presented here is suggestive and theoretically coherent; it is not yet conclusive.

### D. Conditions for Refutation

The hierarchical model would be refuted by any of the following empirical findings: (1) a high-CLI system exhibiting tight coupling between micro-level and macro-level change rates, demonstrating that constitutional architecture does not generate level-specific dynamics; (2) a normative regime whose persistence depends entirely on the individual fitness of its component norms rather than on system-level structural properties; (3) cross-level interactions that do not follow the predicted pattern of damped upward causation in high-CLI systems and facilitated upward causation in low-CLI systems. The model is falsifiable in Popper's sense, and its progressive character in Lakatos' sense depends on its continued ability to generate novel predictions that survive empirical testing.

## IX. Implications for Institutional Design and Reform Strategy

The hierarchical model suggests that reform strategies must be calibrated to the level at which change is sought. Micro-level changes require interventions targeting the HBU mechanism: modifying judicial incentives, restructuring legal education, altering the signals that institutional gatekeepers send. Meso-level changes require structural interventions altering the emergent properties of normative regimes: disrupting internal coherence, weakening institutional embeddedness, or reconfiguring coalition support. Macro-level changes require displacing the system from its basin of attraction, which in high-CLI systems demands either extraordinary political energy or an exogenous shock sufficient to overcome constitutional restoring forces.

This framework explains why incremental reforms so often fail to produce structural change in high-CLI systems. Micro-level modifications, however numerous, do not accumulate into macro-level transformation because they operate within a basin of attraction that actively dissipates perturbations. Argentina's 23 failed reform attempts did not fail because they were individually flawed; they failed because no individual reform could generate sufficient displacement to escape the basin.

A deeper implication emerges from the Henrich-Hanson insight that human preferences are endogenous to cultural evolution. If what individuals value, desire, and believe is itself a product of the institutional environment in which they develop, then law is not merely a mechanism for coordinating pre-existing preferences; it is a mechanism that shapes the preferences it then coordinates. The legislator is not an arbiter of fixed interests but a designer of human nature. This creates a feedback loop of extraordinary theoretical significance: the normative regime shapes the preferences of the actors who then sustain or challenge the regime. In high-CLI systems, this feedback loop is self-reinforcing: the constitutional architecture generates preferences for stability, risk aversion, and institutional continuity among the legal professionals and political actors who operate within it, which in turn deepens the basin of attraction. The HBU mechanism is the transmission channel for this feedback: legal professionals learn to value whatever institutional gatekeepers reward, and institutional gatekeepers reward whatever is compatible with the existing constitutional architecture. The system produces the actors who reproduce the system.

Dennett's distinction between competence without comprehension and comprehension-driven intelligent design applies directly here. Most institutional evolution proceeds through competence without comprehension: norms persist and proliferate through mechanisms their carriers do not understand, just as termites build elaborate castles without architectural blueprints. The aspiration of institutional design is to introduce comprehension into this process, to shift from blind bottom-up evolution to informed top-down design. But the hierarchical model suggests that comprehension-driven reform faces a fundamental obstacle: the emergent properties that generate macro-level stasis are precisely the properties that are least visible to designers working at the micro level. A legislator proposing a labor market reform can see the individual norms being modified but cannot see the basin of attraction that will absorb the modification and restore the equilibrium.

Dennett's concept of 'thinking tools,' the cultural artifacts that enhance our cognitive capacity and enable top-down design, offers a partial solution. The CLI, the IHR, the IEI, and the IusSpace framework itself are thinking tools in Dennett's sense: cognitive instruments that make visible the emergent properties that natural human perception cannot detect. They function in institutional analysis the way microscopes function in biology or telescopes in astronomy: they extend the range of perception to scales that unaided cognition cannot access. Without such tools, institutional reformers are like pre-microscope physicians: they can observe symptoms and attempt treatments, but they cannot see the mechanisms producing the pathology. With the tools developed within this research program, the emergent properties that generate macro-level stasis become, for the first time, measurable and potentially manipulable.

The process that Dennett describes as 'de-Darwinizing,' the gradual transition from blind evolutionary trial-and-error to informed, comprehension-driven design, applies to legal evolution as well. Early legal systems evolved through purely bottom-up processes: customs emerged, persisted if functional, and died if they proved maladaptive. The codification movement of the nineteenth century represented a partial de-Darwinization: legislators attempted to replace blind customary evolution with rational legislative design. But the hierarchical model suggests that codification de-Darwinized only the micro level, replacing customary variation with legislative design. The meso and macro levels continued to evolve through largely blind processes: regimes competing through emergent fitness, constitutional orders persisting or collapsing based on basin dynamics that no designer controlled. The simulation tools that Shapira et al. have demonstrated may enable a further step in de-Darwinization, extending comprehension-driven design to the meso level by modeling the emergent properties of normative regimes before they are enacted. Full de-Darwinization of the macro level, however, would require the capacity to model and manipulate basin depths, a capacity that remains beyond current computational reach.

The February 23, 2026 publication of Shapira et al. (arXiv:2602.20021), demonstrating that institutional simulation tools can model policy effects before implementation, establishes a new threshold for legislative design. If the dynamics described in this Article are correct, pre-deployment simulation must account for hierarchical effects: a reform appearing beneficial at the micro level may be neutralized by meso-level structural properties or reversed by macro-level basin dynamics. The concept of pre-market normative evaluation, grounded in the existing Argentine duty of legislative diligence (Articles 1757-1758 CCyCN), becomes not merely desirable but arguably legally required once simulation tools achieve sufficient predictive accuracy.

The WEIRD/non-WEIRD distinction carries further practical implications. Reform strategies designed in WEIRD institutional environments, where the three tiers are tightly coupled and formal institutions are the primary coordination mechanism, may fail catastrophically when transplanted to non-WEIRD environments where informal kinship-based institutions dominate the coordination landscape. The IusSpace framework indicates that reform vectors must account not only for the distance traveled in formal institutional space but also for the 'cultural friction' generated by the informal institutional substrate. The Latin American pattern of 'se acata pero no se cumple' is not a failure of implementation but a predictable consequence of attempting formal institutional change in environments where informal institutions exercise dominant downward causation on actual behavior.

The hierarchical model also reframes the debate about legal transplants. Watson's famous thesis that legal transplants are the primary mechanism of legal change, and Legrand's equally famous riposte that successful transplants are impossible because law is embedded in culture, can both be understood as partially correct descriptions of dynamics occurring at different hierarchical levels. At the micro level, individual norms transplant with relative ease: a specific contract provision, a particular procedural rule, a doctrinal formulation can move from one system to another and replicate successfully. At the meso level, normative regimes transplant with greater difficulty because their emergent fitness depends on structural properties that may not survive the move to a new institutional environment. At the macro level, constitutional architectures are virtually non-transplantable because they are embedded in basins of attraction shaped by centuries of institutional co-evolution. Watson is right about micro-level transplants; Legrand is right about macro-level ones; both are partially right and partially wrong about the meso level.

A final implication concerns the nature of legal knowledge itself. If legal evolution proceeds through the three-tier hierarchical process described in this Article, then legal knowledge is distributed across levels in a way that no single actor can fully comprehend. Practitioners understand the micro-level dynamics of doctrinal evolution because they participate in HBU directly. Legislators and policy designers may understand meso-level dynamics of regime competition, though often imperfectly. But the macro-level dynamics of constitutional lock-in and basin depth are largely invisible to all actors, operating through emergent properties that no individual designed and no individual controls. This is Dennett's competence without comprehension applied to institutional evolution: the legal system as a whole exhibits extraordinary competence in maintaining its structural integrity and absorbing perturbations, yet this competence is not grounded in any actor's comprehension of how the system works. The system is more like a termite castle than a Gaudi cathedral: its complexity emerges from the aggregated behavior of agents following local rules, not from top-down architectural planning.

This observation carries a note of epistemic humility for would-be institutional reformers. The aspiration to redesign legal systems through rational planning confronts a fundamental limitation: the emergent properties that determine macro-level outcomes cannot be directly manipulated because they arise from structural relationships among components rather than from the components themselves. A legislator can modify a statute; a judge can reinterpret a doctrine; but no actor can directly modify the depth of the basin of attraction in which the constitutional order sits, because that depth is an emergent consequence of the entire architecture, not a parameter that can be adjusted independently. Effective institutional reform, in high-CLI systems, may require not frontal assault on the basin but indirect strategies that gradually alter the structural relationships from which basin depth emerges: changing the amendment procedure, modifying judicial review scope, restructuring the treaty hierarchy. These are not reforms of individual norms but reforms of the architecture that generates the emergent fitness of the constitutional order.

## X. Epistemological Positioning

*This Article adopts an explicitly Lakatosian epistemological position. The Extended Phenotype Theory research program constitutes a progressive research program in Lakatos' (1978) sense: its hard core (legal norms as memes subject to Darwinian selection; institutions as extended phenotypes of competing memeplexes) has remained stable while its protective belt of auxiliary hypotheses has expanded to accommodate new phenomena and generate novel predictions. Popper's (1959) falsifiability criterion provides the demarcation standard: every claim is formulated as a falsifiable hypothesis with specified conditions for refutation. Hull's (1988) model of science as a selection process and Campbell's (1974) evolutionary epistemology provide the meta-theoretical framework for understanding the research program's own evolution as an instance of the Darwinian dynamics it describes. The integration of Gould's hierarchical selection into the EPT framework represents an expansion of the protective belt, not a modification of the hard core: the norm remains the fundamental replicator; what changes is the recognition that selective pressures operating on norms vary qualitatively across hierarchical levels.*

*A note on Kuhn's relevance: the present Article does not claim that evolutionary jurisprudence constitutes a Kuhnian paradigm shift in legal theory, a claim made in an earlier paper within this research program. The Kuhnian framework is useful for understanding the sociology of theoretical competition in legal scholarship, and the anomalies that confront legal positivism and natural law theory are genuine. But the relationship between the evolutionary approach and the traditional paradigms is better understood through Lakatos than through Kuhn. EPT does not make legal* *positivism or natural law theory 'wrong' in the way that Lavoisier made phlogiston theory wrong; it subsumes them as special cases within a more general framework. Legal positivism correctly identifies the social-fact basis of legal validity; EPT explains why those social facts take the forms they do. Natural law theory correctly identifies transcultural normative convergence; EPT explains why convergence occurs (environmental compatibility) without invoking metaphysical moral realism. The relationship is one of theoretical generalization, not revolutionary replacement, and Lakatos' model of progressive research programs captures this relationship more accurately than Kuhn's model of revolutionary paradigm shifts.*

## XI. Conclusion: Toward a Unified Evolutionary Jurisprudence

The Dawkins-Gould debate, which has resisted resolution for four decades in biology, dissolves in the domain of legal evolution. The dissolution is not trivial: it suggests that the Weismann barrier, which makes the biological dispute intractable, is replaced in cultural evolution by a Lamarckian transmission mechanism that paradoxically generates rather than eliminates hierarchical structure. The coexistence of rapid micro-level adaptation with rigid macro-level stasis is not an anomaly but a direct prediction of the model.

This Article's contribution to the Extended Phenotype Theory research program is fourfold. First, it reconciles the replicator-centered foundation of EPT with multilevel selection theory, showing them to be complementary aspects of a single evolutionary process. Second, it connects EPT to broader philosophical traditions through Bunge's systemism, Dennett's universal Darwinism, Godfrey-Smith's population thinking, Henrich's gene-culture coevolution, and Pievani's contingency perspective. Third, it meets Godfrey-Smith’s challenge head-on, arguing that legal norm populations satisfy the criteria for paradigm Darwinian populations more fully than most cultural replicators. Fourth, it generates falsifiable predictions supported by preliminary evidence from four jurisdictions.

The integration achieved here is not eclectic but systematic. Each tradition contributes a specific conceptual tool: Dawkins supplies the replicator concept and the extended phenotype; Gould supplies hierarchical structure, punctuated equilibria, and emergent fitness; Bunge supplies the systemist ontology and CESM; Dennett supplies Design Space and competence without comprehension; Godfrey-Smith supplies the paradigm/marginal population distinction; Henrich supplies gene-culture coevolution and the WEIRD/non-WEIRD bifurcation; Pievani supplies contingency and exaptation. The result is a framework whose emergent properties, fittingly, exceed those of its component traditions.

The research program's trajectory is clear. The 'Institutional Stratigraphy' paper, next in the pipeline, will apply the hierarchical model to construct a prospective fossil record of institutional change using CLI, IHR, and IEI as stratigraphic markers. The integration with Kozlov's formal conditions framework will provide computational infrastructure for testing predictions at scale.

And the expansion of the empirical base will either strengthen the model's accuracy or reveal its boundary conditions.

Several open questions demand attention. First, the precise mechanisms of upward and downward causation between levels need formal specification. The present Article identifies these interactions qualitatively; future work should develop quantitative models, potentially drawing on the evolutionary game theory apparatus already deployed within this research program, to predict when upward causation will propagate and when it will be damped by macro-level basin depth. Second, the IusSpace framework needs extension to incorporate informal institutional dimensions that the current twelve formal dimensions do not capture. The WEIRD/non-WEIRD implementation gap suggests that informal institutions contribute a substantial fraction of the total basin depth in non-WEIRD systems, and this 'dark institutional matter' must be made measurable if the model is to achieve full predictive power. Third, the relationship between the hierarchical model and the CriptoIus project, which proposes blockchain-based infrastructure for legal precedent tracking, needs exploration: if the model is correct, CriptoIus should be designed to track not only individual-norm fitness but also the emergent properties of normative regimes and the basin depths of constitutional orders.

The present Article advances the EPT research program by incorporating what may be its most significant theoretical expansion since the original formulation: the recognition that Dawkins and Gould were both right, in their respective domains, and that the apparent conflict between their frameworks dissolves precisely in the domain, cultural evolution, where the practical stakes are highest. This dissolution is not a compromise or a splitting of differences; it is a genuine synthesis that generates predictions neither framework could generate independently. The hierarchical model predicts the coexistence of micro-level plasticity with macro-level rigidity, a pattern that the gene-centered view cannot explain (because it predicts uniform change rates across levels) and that pure multilevel selection cannot explain (because it does not identify the Lamarckian mechanism that generates the micro-level plasticity). Only the synthesis captures both halves of the empirical pattern.

Gould concluded The Structure of Evolutionary Theory with the observation that the history of life is not a tale of simple progress but of contingent, multilevel, endlessly creative proliferation. The same may be said of the history of law. Legal evolution is not the story of rational design gradually optimizing institutional arrangements; it is the story of blind replicators building elaborate phenotypic expressions, competing at multiple levels of a hierarchy they never intended to create, and generating patterns of stasis and punctuated change that reflect the deep structure of cultural selection. The task of evolutionary jurisprudence is not to replace this process with rational design but to understand it well enough to work with it rather than against it, deploying what comprehension Dennett's thinking tools afford us within a system whose complexity will always exceed our capacity to control it.

The Adam Smith genealogy that connects the intellectual traditions mobilized in this Article deserves a final observation. Smith's insight that complex order can emerge from the uncoordinated actions of self-interested agents, the 'invisible hand,' stands at the origin of two intellectual lineages that the present Article brings together. One lineage runs from Smith through Malthus to Darwin and thence to Dawkins, Dennett, and Gould: the evolutionary tradition that explains biological complexity through blind variation and selective retention. The other runs from Smith directly to Mises, Hayek, and the Austrian economics tradition: the spontaneous order tradition that explains institutional complexity through decentralized coordination. Both traditions recognize that complex adaptive systems exhibit design without designers, competence without comprehension, and order without central planning. The hierarchical model proposed here unites these two Smithian lineages in the domain of law, showing that legal institutions are simultaneously products of cultural evolution (in the Dawkins-Gould sense) and instances of spontaneous order (in the Hayek sense), and that the tension between these two descriptions disappears once the hierarchical structure of the evolutionary process is properly understood.

The practical consequence of this understanding is sobering but liberating. It is sobering because it implies that the structural features of legal systems that frustrate reformers, their rigidity, their resistance to rational design, their tendency to repurpose institutions for functions they were never designed to serve, are not pathologies to be cured but features of a hierarchical evolutionary system operating as evolutionary theory predicts. It is liberating because it suggests that effective institutional intervention is possible, not through the frontal assault of comprehensive reform, but through the indirect strategy of modifying the architectural parameters that generate emergent fitness at the macro level: the amendment rules, the judicial review doctrines, the treaty hierarchies, and the veto player configurations whose interaction determines the depth of the basin of attraction. The hierarchical model does not counsel passivity in the face of institutional dysfunction; it counsels intelligence in the choice of intervention level.

### *AI Disclosure*

*Artificial intelligence tools were used as research support during the preparation of this Article, specifically to locate sources, extract information from the existing literature, and review drafts for consistency. The thesis, analytical framework, theoretical arguments, and all conclusions represent the original intellectual contribution of the author. This is independent research; no AI tool drafted, structured, or provided substantive feedback on the conceptual architecture of the hierarchical selection model proposed herein.*

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